Product Quality Proliferation and Differentiation in Export Markets

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Abstract

Multiproduct firms often export only a portion of their available product range to any given market. This paper analyzes and empirically tests a multi-product, multinational firm's export decisions about product quality proliferation. It considers a duopoly where the multiproduct firm chooses whether to export one or both of its quality levels. The exporting firm chooses both the number and the level of quantities for export. Firms choose their products for export in Stage 1 of the game and compete in prices in Stage 2. Solving for the subgame perfect Nash equilibrium reveals that product proliferation in qualities is profitable if tariffs or trade costs are low enough and quality differentiation is not too large. Lower tariffs or trade costs enable the multi-product firm to steal more market share from the competitor's product. The degree of quality differentiation determines whether the multiproduct firm benefits from exporting an "aggressor product," which is closer in quality to the competitor's product but earns low profit margins. If quality gaps are large, then the "aggressor product" cannot attract enough customers away from the competitor firm to warrant the increased intra-firm competition. Data from the Mexican auto market, which experiences progressive trade liberalization starting in the mid-1990s due to multiple trade agreements, provides empirical support for the relationship between product proliferation in qualities and lower tariffs, lower trade costs, a tariff advantage, and a smaller quality gap. JEL: F10, L13 Keywords: product differentiation, product proliferation, export product choice, multi-product firms, trade liberalization

1 Introduction

Multinational firms are typically also multiproduct firms: Bernard et al. (2007) document that in 2000, U.S. firms that export more than five products accounted for 98 percent of export value. And even though these firms accounted for only 25.9 percent of exporting firms in the U.S., more than half of exporting firms exported more than one product. Firms also frequently offer many quality levels of any given product. The auto industry offers numerous examples. For instance, there are three different trim levels of the Honda Civic available, with the more expensive versions offering more options such as navigation systems, satellite radio, etc. The many versions of the BMW 3 Series available vary in engine size, cylinders, horsepower, torque, and fuel capacity. The practice is common throughout the industry: multiple versions of each model are available, allowing variable qualities of the same product through different engines, options, and even body styles.

While multinational firms often export multiple products or versions of the same product, the range of available products is often bigger than what is sold in any single export market. For example, automobile manufacturers often sell only cars with large, powerful engines and automatic transmissions in Mexico while they sell cars with smaller, often diesel, engines in Europe. Similarly, not all car models from any given manufacturer, or all versions of any given car model, are sold in the United States. In 2009, for example, three Honda Civic models were sold in the U.S. (the DX, LX, and EX trims) but only two were sold in Mexico (EX and LX only).[1] In the same year, there was little overlap in the versions of the BMW 5 Series that were sold in Mexico and the U.S.: the 528, the 535, and the 550 were sold in the U.S., while consumers in Mexico could choose instead from the 525, the 530, and the 550. Consumers in England, however, could choose from six different versions of the BMW 5 Series (the 520, 523, 525, 530, 540, and 550).[2] Kitchen appliances also come in many quality levels, as do computers, power tools, and stereo equipment. Regardless of the industry or product category, often only a portion of the available versions are exported to any given market.

Firms may limit the product range available in a given market for a number of reasons. A particular market might be too small to support the full range of versions; consumers may be unable to afford the highest quality levels available if income levels are low; or high income levels may eliminate the need for the lowest quality products. Given the market conditions, a firm must determine both how many versions of each product to export to a particular market and which specific versions to export.

As Figure 1 suggests, tariff levels may affect these decisions: if high tariff levels effectively limit a firm's

[1] Occasionally, a make or model will be renamed for sale in a foreign country without any substantive changes to the product itself. These are not considered different products.

[2] These were just the petrol engines; there were an additional 4 diesel versions available.

competitiveness, the firm may export only one version of a product. The figure shows, for each make and model of car in a given year, how many versions of that product were exported to the Mexican market.[3] High tariffs may constrain demand for the product to the extent that segmentation of customers is not profitable. Lower tariffs, on the other hand, may encourage product proliferation. In addition to its own tariffs, a firm will also naturally consider the characteristics and prices of competing products when making its decision about which versions of a product to export to a foreign market.

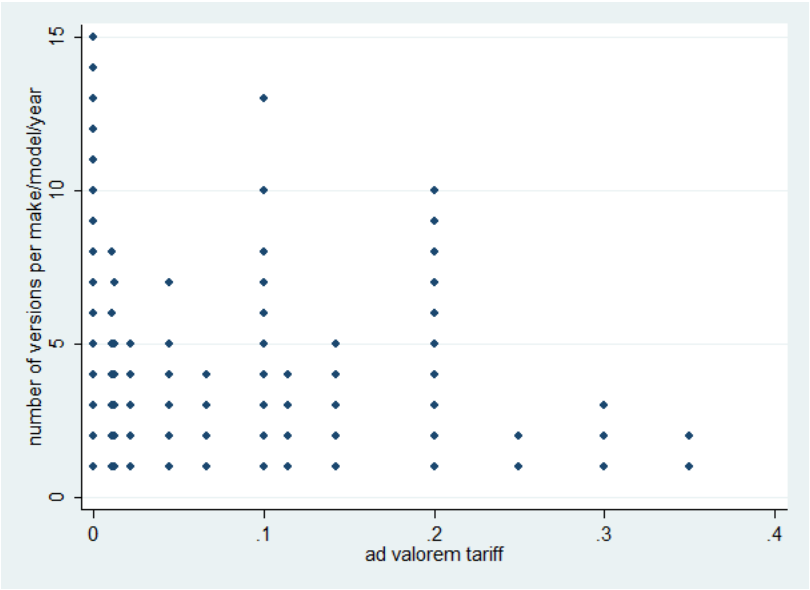

Figure 1: Plot of the number of versions for a make/model/year against the tariff

This paper focuses on an exporting firm's choice of how many and which quality levels to export to a foreign market. In particular, it addresses the question of whether lower tariffs make quality proliferation profitable for the firm in the presence of competition from another firm's product, and how quality and cost parameters affect this decision. That is, if the importing country reduces the tariff rate applied to its goods, does a firm react with product proliferation in qualities? The focus on how many versions the firm exports is a key contribution of the paper. Instead of choosing only the quality level, the firm must make choices in two dimensions, quality level and quality proliferation. The paper is thus distinct from much of the previous literature in that is does not impose an *ad hoc* one-product limit on the firm, instead allowing for the (average) quality level of imports to change through proliferation of qualities.

The paper is similar in spirit to the literature on third degree price discrimination in oligopoly. While that literature (see for example Holmes 1989, Schulz 1999, and Dastidar 2006) considers single-product firms competing in two separate markets, this paper considers multiproduct firms competing in a single (export) market. But in both cases, a central question is whether price discrimination is profitable (in these previous

[3] Each version has a unique combination of trim level, body style, and transmission. See Section 4.1 for detailed discussion.

papers the discrimination is between independent markets, while in this paper the price discrimination is achieved by using quality to distinguish between consumers in a single market). In all cases, results show that price discrimination may not always be profitable. However, in the current paper, the relationship of the single-quality (i.e. no discrimination) price to the prices when both qualities are offered depends on the ordering of qualities (and thus on the existence of different qualities). Furthermore, since the current paper considers multiproduct firms with asymmetric, vertically differentiated products in protected export markets, the profitability of price discrimination can additionally be tied to the added features of the model (the quality difference and the level of protection).

While previous papers have considered the impact of tariffs on the quality of imports, they assume that each exporting firm has either a fixed number of products or a continuum of products. In one of the earliest papers in this literature, Falvey (1979) analyzes the impact of tariffs and quotas when exactly two different quality levels of the same good are imported. Papers by Krishna (1987, 1990a), Das and Donnenfeld (1987, 1989), and Schmitt (1995) also study the impact of restrictive trade policies on the quality of imports. However, most of these papers (Krishna 1987, Das and Donnenfeld 1987) consider only the monopoly case, while those papers that analyze a duopoly consider only single-product firms (Das and Donnenfeld 1989). This paper, on the other hand, considers a duopoly with a multiproduct firm, where goods are vertically differentiated and qualities are asymmetric. Critically, in this paper I allow the multiproduct firm to choose how many quality levels to export rather than limiting the firm to a single product.

The current analysis is also closely related to the product line rivalry literature, and in particular its focus on product selection by multiproduct firms.[4] Brander and Eaton (1984) use the terms "market segmentation" and "market interlacing" to describe the potential configurations of firms in a duopoly in which each firm produces two horizontally differentiated products.[5] They focus on a case in which each firm chooses a pair of products (out of four available products) to produce and then competes in quantities; both segmentation and interlacing are possible outcomes, although the latter is excluded if product line choices are sequential rather than simultaneous.[6] If the decision of how many products to produce is endogenized, as in Maritnez-Giralt and Neven (1988), then firms choose to produce only one horizontally differentiated product each rather than two in order to relax price competition.[7] And when the model is modified to allow for asymmetric costs, then interlacing is also an equilibrium outcome even when firms choose product lines sequentially (Chen and

[4] This is distinct from the multiproduct firms literature that focuses on economies of scope; I abstract from economies of scope.

[5] Market segmentation describes the case when a firm's products are close substitutes for each other but not for the rival firm's products; market interlacing describes the reverse, when the firm's products are close substitutes for the rival's products but not for each other.

[6] In a note on the paper, Wernerfelt (1986) finds that if fixed costs are high firms will offer only one product each. Whether the products are differentiated or standardized depends on the degree of homogeneity of consumer tastes.

[7] This result holds for both a Hotelling model and a circle model. In contrast, Verboven (1999) considers a symmetric duopoly where each firms offers both a high-quality and a low-quality product.

Chen (2011)).[8]

The paper is formally close to Neven and Thisse (1990), who analyze a duopoly where products are both horizontally and vertically differentiated and find that the firms will maximally differentiate along exactly one dimension. Gilbert and Matutes (1993) use a similar setup to study an oligopoly with differentiation in two dimensions. In contrast to Neven and Thisse (1990), however, they allow each firm to produce up to two vertically differentiated products, but they assume that the firms are symmetric and that production exhibits economies of scope. The current model incorporates only vertical differentiation but assumes that the available quality levels are asymmetric and that cost is increasing in quality. The quality levels are also discrete and finite instead of being on a continuum. The choice of quality and variety from a continuum are more relevant to large (including world-wide) markets and the R&D stage of the product life-cycle, while the choice from an existing product line with discrete versions is more appropriate to small export markets. The difference is also one of long-term versus a medium-term analysis: in the long run, firms can adjust the quality of each product by making investments in research, development, and design activities. In the medium term, however, the firm's investment is fixed. Instead, it chooses which products to sell in each market, and then sets a price for each product.

In a recent development that combines these two strands of inquiry, Bernard, Redding, and Schott (2011) develop a model of multiproduct firms that trade internationally.[9] They find that there is a correlation between the intensive and extensive export margins. Firms with more ability have both greater product scope and export more of any given product. As a result of trade liberalization, firms drop the products in which they have the least expertise, so that trade liberalization reduces product scope but increases average expertise. The paper considers the impact of sudden globalization (rather than marginal trade liberalization). In addition, consumers are all symmetric and purchase multiple products due to a love of variety rather than a willingness to pay for quality.[10]

This paper analyzes the response of multiproduct firms to a reduction in trade costs using a duopoly model. Products are differentiated in quality. The available levels of quality are discrete and fixed, representing the notion that multinational firms often choose which versions (qualities) to export to a given market from their existing product portfolios rather than developing a new product quality for each export market.[11] One of the firms has three possible strategies: export only its lower quality product, export only

[8] See also Canoy and Peitz (1997) for a model with asymmetries among firms: there are two high-quality goods that are maximally horizontally differentiated and only one low-quality good.

[9] Firms are heterogeneous in two dimensions, ability and product expertise.

[10] Ries (1993) also considers multiproduct firms. The firms compete in quantity, have asymmetric cost functions, and produce all products that are profitable. Under quantity constraints, he finds that the firm initially producing the lower quality goods has no incentive to upgrade quality; it may, however, be optimal for both firms to produce a lower quality than prior to the imposition of a quantity constraint. Although this finding contradicts the quality-upgrading results of many single-good firm models, it does still find that quantity constraints may lead to an adjustment in the quality of goods being produced.

[11] While the central question is similar to one of the questions in Bernard et al. (2011), the approach is quite different. Bernard

its higher quality product, or export both products. The central question is whether, as the tariff rate falls, the firm continues to sell only one of its quality levels or chooses instead to export two of them.

The multiproduct firm's optimal strategy depends on multiple factors, including the relative ordering of quality levels, the distance between both quality levels and varieties, the size of the tariff preference, the size of the tariff itself, and the cost of quality. Under some conditions, as tariffs fall the two-quality strategy profits grow faster than the single-quality strategy profits. If tariffs far fall enough, profits under the two-quality strategy overtake those under the single-quality strategy, and the firm chooses to export both of its quality levels, instituting quality discrimination among its own customers.

Analysis of the model shows that if the firm's available quality levels are close enough to each other and to the quality level of the competing product, then as tariffs fall the firm benefits from selling both of its quality levels. In this case, the product that is closer in quality to the competitor's product acts as a market share aggressor. However, if the gap between the quality level of the firm's own products and that of its competitor is too large, then this aggressor product is not beneficial, as it does not acquire sufficient market share from the competitor yet still introduces intra-firm competition. If the firm chooses to introduce an aggressor product to the market, that product will compete not only with the competitor's product but also with the firm's other product. When qualities are close, the beneficial aggressor effect outweighs the harmful effect of intra-firm competition. When quality levels are too far apart, competition is weak and the firm is better off selling just one of its products rather than introducing a second product. In this case, the aggressor effect is minimal; instead, the primary impact of the aggressor product is to increase competition for the firm's other product.

In order for profits from the two-quality strategy to exceed those of the single-quality strategy as the tariff falls, the firm's tariff rate must also be lower than its competitor's. If the firm is at a tariff disadvantage, then a decrease in the tariff-inclusive consumer price will increase market share only a little. The product is competitively disadvantaged because it is still subject to a relatively high tariff. If the firm has a tariff advantage, however, as its tariff falls further, the tariff-inclusive price (paid by consumers) is not only lower than it was at a higher tariff rate, is also has a competitive advantage due to the lower tariff rate. This allows the firm's combined market share to increase fast enough for profits from the two-quality strategy to outpace profits from the one-quality strategy. In this case, then, when the firm has a tariff advantage and needs a buffer product, as tariffs fall the firm will choose to export both of its quality levels.

et al. consider only horizontal differentiation of products; a second dimension of differentiation is in the ability or productivity of firms. The firms also produce a continuum of products, so that the average variety is increased with trade liberalization. Furthermore, consumers in Bernard et al. have CES preferences and buy more than one good in a given industry.Products in the present model are differentiated in quality, consumers purchase only one good (e.g. consumer durables), and the products are discrete. Trade liberalization may lead to an increase or a decrease in the average quality of products, depending on the ordering of product qualities.

Data from the Mexican auto market provides empirical support for the hypothesis that trade liberalization induces product proliferation. Using the number of engine and trim levels at the make-model-year level to represent the number of quality levels a firm exports, I find that tariffs have an economically and statistically significant effect on the number of versions of a product that are offered for sale. Evidence on the effect of quality differentiation on product proliferation is mixed. While the effect of some quality measures (engine size) is significant, other quality attributes (cost per mile and physical size) do not have a significant effect on the number of versions offered for sale.

The rest of the paper is organized as follows: Section 2 sets up the oligopoly model and considers parameter bounds, while 3 finds the optimal price and product configuration in equilibrium. Section 4 introduces the data and provides background on the automobile market in Mexico. Section 5 presents the results of empirical analysis, and the final section concludes.

2 Duopoly with Multiple Qualities

There are two firms, $n = \{1, 2\}$, each producing different qualities of a good. Each firm must choose which products to export to a given market. To simplify, Firm 2 has available only one quality of its product, $q = q_2$; the other firm (Firm 1) has available two discrete qualities of its product, $q_j \in (0, \infty), j = \{h, l : q_h > q_l\}$. The quality levels depend on global market research and product engineering constraints and are set prior to choosing the products to be sold in each individual export market. That is, in the current setup, in order to focus on the proliferation of product qualities, I fix the portfolio of product qualities available for production and export.

I assume that the product qualities are asymmetric: $q_2 \neq q_h \neq q_l$. I also assume, without loss of generality, that $q_h > q_l$. Given these assumptions, there are three possible cases to analyze:

- Case 1: Firm 2 produces the highest of the three qualities ($q_2 > q_h > q_l$),
- Case 2: Firm 2 produces the lowest of the three qualities ($q_h > q_l > q_2$), or
- Case 3: Firm 2 produces the median quality ($q_h > q_2 > q_l$).

The marginal cost of producing a product, $c(q)$, is constant in quantity but increasing in quality. Specifically, I assume that the marginal cost is linear in quality, $c(q_j) = k \times q_j, k > 0$. This reflects the notion that a higher quality product is likely to require more expensive materials and production process and more extensive quality control.[12] Finally, each product is also subject to *ad valorem* tariffs levied by the importing

[12] If $k = 0$ so that the marginal cost is normalized to zero regardless of product quality, then there is no reason for Firm 1 to export both products in Cases 1 and 2.

country, $t_n, t \geq 0$. Given the competitive configuration, costs, and tariffs, Firm 1 must decide whether to export one or both of its available qualities to a given market when tariffs are set at t_n.

2.1 Firms

The firms play a 2-stage, non-cooperative game. In Stage 1, each firm chooses its product mix. In particular, Firm 1 can choose whether to export one or both of its products to the market in question. In Stage 2, the firms compete in prices. The subgame perfect Nash equilibrium gives the firms' equilibrium choices. As is typical, the game is solved by backward induction.

This entails first finding the equilibrium prices in the final stage of the game for any product configuration chosen in the first stage. The equilibrium of the entire game is then the pair of strategies that generates the greatest profit for each firm, given the product mix and prices chosen by the other firm. That is, a firm's strategy, s_n, specifies both the products it exports and a price vector for each possible product configuration.[13] Firm 1 may choose to export only its high quality good, only its low quality good, both goods, or no goods. Since Firm 2 has only one good available, it chooses whether to export or not export. I restrict the analysis to a duopoly and therefore focus only on those equilibria in which each firm exports at least one product.

2.1.1 Equilibrium Prices and Profits

The profit generated by exporting quality j are given by $\pi_j = [p_j - c_j] \times D_j(p_j, p_{-j}, q_j, q_{-j}, t_j, t_{-j})$, where D_j is the demand for the good of quality q_j. Demand for a particular product quality depends on the price set by the firm for that quality, the tariff rate, the prices and tariff rates of products with adjacent qualities, and the difference between the quality of the product and the quality of adjacent products.

In stage 2, given the products that are available, each firm chooses its price(s) in order to maximize profits, taking the product configuration and the other firm's prices as fixed:

$$\max_{p_j, j \in J} \sum_{j \in J} [p_j - c_j] \times D_j(p_j, p_{-j}, q_j, q_{-j}, t_j, t_{-j}),$$

where $J_1 \in \{h, l, hl\}$ and $J_2 = 2$ is the set of qualities that the firm has chosen to export.

[13] Formally, Firm 1's strategy set S_1 is $\{h, l, hl, 0\} \times \{p_1(\cdot) : \{h, l, hl, 0\} \times \{2, 0\} \to [0, \infty)^g, g \in \{0, 1, 2\}\}$ and Firm 2's strategy set S_2 is $\{2, 0\} \times \{p_2(\cdot) : \{h, l, hl, 0\} \times \{2, 0\} \to [0, \infty)^h, h \in \{0, 1\}\}$, where $\{p_j(\cdot) : \{h, l, hl, 0\} \times \{2, 0\} \to [0, \infty)\}$ is the space of functions that map from the products that each firm chooses in Stage 1 to the price for each product.

2.1.2 Firm 1's Problem

Given the assumption of duopoly (i.e. that Firm 2 chooses to export), Firm 1 must decide in Stage 1 on its product mix. Its maximization problem in Stage 1 is therefore[14]

$$\max \left\{ \max_{p_h} (p_h - c_h) \times D_h, \ \max_{p_l} (p_l - c_l) \times D_l, \right.$$
$$\left. \max_{p_h, p_l} (p_h - c_h) \times D_h + (p_l - c_l) \times D_l \right\}.$$

Letting $\pi(s_n, p, q)$ be the sum of variable profits for firm n from selling its product mix at optimal prices as given by strategy s_n, Firm 1's problem can be rewritten as

$$\max_{s_1 \in S_1} \pi(s_1, p, q),$$

where S_1 is Firm 1's strategy set as defined in Section 2.1 (see footnote 13). This says that Firm 1 will choose the strategy such that the maximum profit of that strategy is greater than the maximum profit of any other strategy, given the price and quality of the competing product.

2.2 Consumers

Consumers are characterized by a parameter θ, where θ is the consumer's valuation of quality.[15] There is a unit mass of consumers distributed uniformly along the unit interval. Each consumer, i, gets linear utility

$$u_{i,j} = \theta_i q_j - (1 + t_n) p_j + K \tag{1}$$

from purchasing a good of quality j at price p_j. Each consumer buys only one good, naturally the good that delivers the greatest non-negative utility given the consumer's own characteristics and the quality and price of the available products. K is a constant assumed to be big enough such that every consumer has positive utility from at least one variety-quality combination (see for example Neven and Thisse, 1990).

Each pair of adjacent qualities is associated with a marginal consumer, who is just indifferent to purchasing the good of (higher) quality q_j and the good of (lower) quality $q_{j'}$. This quality value of this marginal consumer is designated $\theta_{j,j'}$, which can be expressed as

$$\theta_{j,j'} = \frac{(1 + t_j) p_j - (1 + t_{j'}) p_{j'}}{q_j - q_{j'}}. \tag{2}$$

[14] The dependence of demand on quality levels, prices, and tariffs is suppressed for ease of notation.
[15] A consumer with a higher θ cares more about quality; θ is thus also indicative of income level.

The indifferent consumer determines the market share boundaries for each good. Consumers with $\theta > \theta_{j,j'}$ will always prefer the higher quality good, q_j, while consumers with $\theta < \theta_{j,j'}$ will prefer the lower quality good. It should be noted that the marginal consumer need not always exist: it may be the case that all consumers prefer one good over the other.[16]

2.3 Demand

As stated in Section 2.2, the indifferent consumer determines the boundaries of demand for each product being offered. For example, in Case 3 (where $q_h > q_2 > q_l$), the consumer with quality valuation $\theta_{h,2}$ is exactly indifferent between the good of the higher quality and higher price, q_h, and the competing good of lower quality and lower price, q_2. Those consumers with a higher quality valuation will strictly prefer q_h while those with a lower quality valuation will strictly prefer q_2.[17] Thus, if the firm chooses to sell only its higher quality product in Case 3 (so that q_l is not an option), then demand for each product is as follows:

$$\begin{aligned} D_h(\cdot) &= 1 - \theta_{h,2} \equiv 1 - \hat{\theta} = 1 - \frac{(1+t_1)p_h - (1+t_2)p_2}{q_h - q_2} \\ D_2(x) &= \theta_{h,2} - 0 \equiv \hat{\theta} - 0 = \frac{(1+t_1)p_h - (1+t_2)p_2}{q_h - q_2}. \end{aligned} \tag{3}$$

where $\theta_{h,2} \equiv \hat{\theta}$ is defined as in equation (2).

If only the lower quality is sold, demand is similarly derived. Defining $\tilde{\theta} \equiv \theta_{2,l}$ as quality valuation of the consumer who is indifferent between q_l to q_2, when Firm 1 exports only its low-quality good the demand expressions are

$$\begin{aligned} D_l &= \theta_{2,l} \equiv \tilde{\theta} = \frac{(1+t_2)p_2 - (1+t_1)p_l}{q_2 - q_l} \\ D_2 &= 1 - \theta_{2,l} \equiv 1 - \tilde{\theta} = 1 - \frac{(1+t_2)p_2 - (1+t_1)p_l}{q_2 - q_l}. \end{aligned}$$

[16] The constant term in the utility function, which ensures that every consumer buys something, sets up some asymmetric incentives. In general, each producer has opposing incentives with regard to setting the price: a lower price attracts more customers, but a higher price generates more profit per sale. However, in the case when every consumer buys something, the lowest quality good is affected by only one side of this. In particular, if the constant is assumed big enough so that every consumer buys something, then it is guaranteed that the consumers with the lowest quality valuations will purchase the lowest quality good rather than abstaining from any purchase. The firm that produces this product therefore has no incentives to lower its price to attract those customers. In this case, the producer only has an incentive to raise the price in order to increase the profit per sale (as long as the price isn't so high that the higher quality good is preferred). However, this assumption is made for analytic tractability.

[17] Naturally, if the qualities are ordered so that $q_2 > q_h$ as in Case 1, then consumers with higher quality valuations will prefer q_2 and vice versa.

And when Firm 1 exports both its high- and low-quality goods in Case 3, then demand for each good is[18]

$$\begin{aligned} D_h &= 1-\theta_{h,2} \equiv 1-\hat{\theta} = 1-\frac{(1+t_1)p_h-(1+t_2)p_2}{q_h-q_2} \\ D_l &= \theta_{2,l} \equiv \tilde{\theta} = \frac{(1+t_2)p_2-(1+t_1)p_l}{q_2-q_l} \\ D_2 &= \theta_{h,2}-\theta_{2,l} \equiv \hat{\theta}-\tilde{\theta} = \frac{(1+t_1)p_h-(1+t_2)p_2}{q_h-q_2} - \frac{(1+t_2)p_2-(1+t_1)p_l}{q_2-q_l}. \end{aligned} \tag{4}$$

2.3.1 Bounds on Parameters

These expressions for demand, however, apply only when the demand for each product is positive, or when the price of the higher (lower) quality product is not too high (low). This will hold for each possible strategy when the following condition is met in case 3:[19]

$$0 < \tilde{\theta} < \bar{\theta} < \hat{\theta} < 1.$$

Recall that the optimum prices under each strategy are themselves functions of demand, but the demand expressions hold only under these conditions, which depend on the relationship between prices. These conditions can then be expressed in terms of the parameters of the model, that is, the quality levels, tariffs, and cost. If these conditions are not met, then demand will not be positive for at least one of the qualities, in which case the problem reduces to one of either monopoly or standard duopoly (with one quality each).[20] For example, suppose that Firm 1 is selling only its high-quality good. The price of the high-quality good may be so high that the quality is not high enough to override consumers' dis-utility from paying a higher price. In this case, the demand for the high-quality good will be 0, leaving Firm 2 as the monopolist in the market.

I use the following steps to find the parameter ranges that generate positive demand for each product for each strategy:

1. Solve for equilibrium prices (under each strategy in each Case), assuming that demand is positive.
2. Identify the constraints on prices for positive demand.
3. Substitute equilibrium prices into price constraints to find the parameter restrictions under which equilibrium prices satisfy the price constraints.
4. Verify that the equilibrium prices are indeed optimal.

[18] Demand is defined analogously in Cases 1 and 2, letting $\bar{\theta} \equiv \theta_{h,l}$ define the consumer who is indifferent between the two qualities–in this case adjacent–offered by Firm 1.

[19] The analogous condition for Case 1 is $0 < \bar{\theta} < \tilde{\theta} < \hat{\theta} < 1$ and for Case 2 it is $0 < \tilde{\theta} < \hat{\theta} < \bar{\theta} < 1$.

[20] Since the focus is on market with multiple firms, I assume that parameters are such as to exclude the monopoly case.

I adopt the following notation for the remainder of the paper:

- For Case 1, $q_2 - q_h \equiv a$ and $q_2 - q_l \equiv b$, so that $a < b$ and $b - a = q_h - q_l > 0$
- For Case 3, $\rho \equiv \frac{(q_h - q_2)(q_2 - q_l)}{q_h - q_l}$.

The variable ρ, used in analyzing Case 3, can be rewritten as $\frac{(q_h - q_2)(q_2 - q_l)}{(q_h - q_2) + (q_2 - q_l)}$, showing that it is the ratio of the product of quality differences to the sum of quality differences. It increases as the gap between the highest and lowest qualities (q_h and q_l) increases; the impact of a change in the middle quality (q_2) is ambiguous, depending on whether it is closer to the highest or lowest quality.

As noted, I first derive the optimal price for each product, *assuming* positive demand, by solving the first-order conditions of the profit functions, for each strategy in each case.[21] Table 1 gives these prices.

[21] Recall that the cost of quality is linear, $c_j = kq_j$, so that the profit per unit is $p_j - kq_j$.

	p_h	p_l	p_2
Firm 1's strategy	Case 1 ($q_2 > q_h > q_l$)		
Sell q_h only	$\frac{(q_2-q_h)+2(1+t_1)kq_h+(1+t_2)kq_2}{3(1+t_1)}$	n/a	$\frac{2(q_2-q_h)+(1+t_1)kq_h+2(1+t_2)kq_2}{3(1+t_2)}$
Sell q_l only	n/a	$\frac{(q_2-q_l)+2(1+t_1)kq_l+(1+t_2)kq_2}{3(1+t_1)}$	$\frac{2(q_2-q_l)+(1+t_1)kq_l+2(1+t_2)kq_2}{3(1+t_2)}$
Sell both q_h and q_l	$\frac{(q_2-q_h)+2(1+t_1)kq_h+(1+t_2)kq_2}{3(1+t_1)}$	$\frac{2(q_2-q_h)+3(1+t_1)kq_l+(1+t_1)kq_h+2(1+t_2)kq_2}{6(1+t_1)}$	$\frac{2(q_2-q_h)+(1+t_1)kq_h+2(1+t_2)kq_2}{3(1+t_2)}$
	Case 2 ($q_h > q_l > q_2$)		
Sell q_h only	$\frac{2(q_h-q_2)+2(1+t_1)kq_h+(1+t_2)kq_2}{3(1+t_1)}$	n/a	$\frac{(q_h-q_2)+(1+t_1)kq_h+2(1+t_2)kq_2}{3(1+t_2)}$
Sell q_l only	n/a	$\frac{2(q_l-q_2)+2(1+t_1)kq_l+(1+t_2)kq_2}{3(1+t_1)}$	$\frac{(q_l-q_2)+(1+t_1)kq_l+2(1+t_2)kq_2}{3(1+t_2)}$
Sell both q_h and q_l	$\frac{3q_h+q_l-4q_2}{6(1+t_1)}$ $+\frac{3(1+t_1)kq_h+(1+t_1)kq_l+2(1+t_2)kq_2}{6(1+t_1)}$	$\frac{2(q_l-q_2)+2(1+t_1)kq_l+(1+t_2)kq_2}{3(1+t_1)}$	$\frac{(q_l-q_2)+(1+t_1)kq_l+2(1+t_2)kq_2}{3(1+t_2)}$
	Case 3 ($q_h > q_2 > q_l$)		
Sell q_h only	$\frac{2(q_h-q_2)+2(1+t_1)kq_h+(1+t_2)kq_2}{3(1+t_1)}$	n/a	$\frac{(q_h-q_2)+(1+t_1)kq_h+2(1+t_2)kq_2}{3(1+t_2)}$
Sell q_l only	n/a	$\frac{(q_2-q_l)+2(1+t_1)kq_l+(1+t_2)kq_2}{3(1+t_1)}$	$\frac{2(q_2-q_l)+(1+t_1)kq_l+2(1+t_2)kq_2}{3(1+t_2)}$
Sell both q_h and q_l	$\frac{3(q_h-q_2)+\frac{(q_h-q_2)(q_2-q_l)}{q_h-ql}}{6(1+t_1)}$ $+\frac{3(1+t_1)kq_h+2(1+t_2)kq_2+(1+t_1)kq_2}{6(1+t_1)}$	$\frac{\frac{(q_h-q_2)(q_2-q_l)}{q_h-ql}}{6(1+t_1)}$ $+\frac{3(1+t_1)kq_l+2(1+t_2)kq_2+(1+t_1)kq_2}{6(1+t_1)}$	$\frac{(q_h-q_2)(q_2-q_l)}{3(1+t_2)(q_h-q_l)}$ $+\frac{2(1+t_2)kq_2+(1+t_1)kq_2}{3(1+t_2)}$

Table 1: Optimal Prices

Since these prices are calculated from profit functions that *assume* positive demand, it is necessary to verify that the optimal prices found by assuming that demand is positive are indeed within the constraints on prices for positive demand. For example, if Firm 1 chooses to sell only its high-quality good in Case 3, then there is some p_h^* such that for any $p_h < p_h^*$, Firm 2's demand will be 0. Similarly, there is some p_h^{**} such that for any $p_h > p_h^{**}$, Firm 1's demand will be 0. Only when $p_h^* < p_h < p_h^{**}$, is the demand for both Firm 1 and Firm 2 positive. This is analogous to saying that in order for demand for both Firm 1 and Firm 2 to be positive when Firm 1 sells only its high-quality good, it must be true that $0 < \hat{\theta} < 1$. Similar constraints apply when Firm 1 sells only its low-quality good, and in Cases 1 and 2.

When Firm 1 sells both of its goods, additional constraints apply to ensure that demand for both the high-quality and the low-quality good is positive: $\tilde{\theta} < \bar{\theta} < \hat{\theta}$. This ensure both that Firm 2 has positive demand, and that transitivity is satisfied. Substituting optimal prices into all the constraints produces the parameter ranges that generate positive demand; they are given in Table 2.[22]

Finally, it is necessary to show that the equilibrium prices under the assumption of positive demand are indeed optimal. For example, I need to rule out the possibility that when selling only its low-quality good in Case 1 Firm 1 could earn a higher profit by lowering its price (so that Firm 2 has zero demand at some variety preferences) than by charging the optimal price that still generates positive demand. Such possibilities are ruled out because each Firm's profit function is quasiconcave, so that the optimal price assuming positive demand is also the global optimum.[23] In the single-good strategy, Firm 1 has three options: it can set a price so low that demand for Firm 2's product is 0; it can set a price so high that demand for its own product is 0; or it can set a price such that both firms have positive demand (as assumed above). Define p' as the highest price for Firm 1 that leaves Firm 2 with no demand and p'' as the lowest price for Firm 1 that leaves Firm 2 with all demand. The assumption of positive demand implies that $p' < p_l < p''$. Clearly, Firm 1's profit is decreasing in p_l for any $p_l < p'$ and profit is 0 for any $p_l > p''$. It is also easy to see that Firm 1's profit is strictly concave for $p' < p_l < p''$. Furthermore, Firm 1's profit function is continuous (although not necessarily differentiable) at both p' and p'', so that Firm 1's profit function is quasiconcave and the optimal price is the global optimum given the parameter bounds identified above.[24] The same method of proof applies to the other cases and to Firm 2.

[22] Although there are two constraints in the case of exporting either q_h or q_l alone and four if exporting both of them, when expressed in the primitives of the model these all reduce to three restrictions for all strategies in Cases 1 and 2. This is because in Case 1 (Case 2), the constraints when selling only the high-quality (low-quality) product are always tighter, and therefore imply the constraints when selling only the low-quality (high-quality) good. Furthermore, the constraints when selling both goods imply some of the constraints in the single-good case. In Case 3, however, there remain 7 restrictions on the parameters.

[23] The proof in the single-good case is a simpler version of the proof of Proposition 1 in Neven and Thisse (1990).

[24] If the parameter bounds are violated, then at the optimal prices that assume positive demand for both firms, demand is 0 and per-unit profits are negative either for Firm 1 or Firm 2. In this case, lowering the price may result in positive demand but will lead to negative per-unit profits, while raising the price will still result in no demand. The firm therefore stops exporting, leaving the other firm as the monopolist. This situation can occur when one firm's tariff is too high relative to the other firm's tariff rate.

Parameter Restrictions for Positive Demand at each Location
Case 1 ($q_2 > q_h > q_l$)
$\frac{q_2}{q_h} > \frac{2-k(1+t_1)}{2-k(1+t_2)}$ $\frac{q_2-q_h}{q_2} > \frac{2k(t_1-t_2)}{2-k(1+t_1)}$ $\frac{k(1+t_1)}{2} > 0$
Case 2 ($q_h > q_l > q_2$)
$\frac{q_l}{q_2} > \frac{1+k(1+t_2)}{1+k(1+t_1)}$ $\frac{q_l-q_2}{q_2} > \frac{2k(t_1-t_2)}{1+k(1+t_1)}$ $k(1+t_1) < 1$
Case 3 ($q_h > q_2 > q_l$)
$\frac{q_h}{q_2} > \frac{1+k(1+t_2)}{1+k(1+t_1)} > \frac{q_l}{q_2}$ $\frac{q_h}{q_2} > \frac{2-k(1+t_2)}{2-k(1+t_1)} > \frac{q_l}{q_2}$

Table 2: Parameter Restrictions for Positive Demand

Comparative statics show that a bigger difference in qualities increases the set of parameters that satisfy the constraints.

Proposition 1. *For Cases 1 and 2, a bigger difference between Firm 1's qualities and Firm 2's quality ($|q_h - q_2|$ and $|q_l - q_2|$) increases the range of parameters for which positive demand obtains for all goods. For Case 3, as long as $k(1+t_n) \leq 1$, a bigger difference between Firm 1's own qualities ($q_h - q_l$) increases the range of parameters for which market-wide demand obtains.*

If the difference in quality levels is small then the resulting difference in price will also be small (since the quality difference is not big enough to justify a big difference in price). The firms compete intensely in price and try to undercut each other, which may result in one firm setting the price so low that it captures the entire market. A big difference in quality levels, on the other hand, reduces the competition between firms. Both firms can set higher prices, thereby earning higher per-unit profits, and neither is driven out of the market.

The cost parameter and tariff rates also matter to the tightness of the constraints. If the tariff advantage of one firm is too big, then the price that consumers face is so low relative to the other firm's price, that the other firm is priced out of the market. Hence, a smaller tariff advantage increases the set of parameters that satisfy constraints. In addition, a high cost of quality can exacerbate a tariff disadvantage and price the firm with the disadvantage out of the market. A higher cost necessitates a higher price, and when this higher price is magnified by a higher tariff, then the resulting price seen by consumers may be so high that there is no demand for the good given its quality.

I limit the remainder of the paper to those parameter sets that satisfy the constraints.

3 Variable Profits and Optimal Strategies

Using the prices in Table 1, Firm 1's profits at optimal prices for each of its three possible strategies are given in Table 3.

	Firm 1's profit
Firm 1's strategy	Case 1 ($q_2 > q_h > q_l$)
Sell q_h only	$\frac{[q_2-q_h-k(1+t_1)q_h+k(1+t_2)q_2]^2}{9(1+t_1)(q_2-q_h)}$
Sell q_l only	$\frac{[q_2-q_l-k(1+t_1)q_l+k(1+t_2)q_2]^2}{9(1+t_1)(q_2-q_l)}$
Sell both q_h and q_l	$\frac{[q_2-q_h-k(1+t_1)q_h+k(1+t_2)q_2]^2}{9(1+t_1)(q_2-q_h)} + \frac{k^2(1+t_1)}{4}(q_h-q_l)$
	Case 2 ($q_h > q_l > q_2$)
Sell q_h only	$\frac{[2(q_h-q_2)-k(1+t_1)q_h+k(1+t_2)q_2]^2}{9(1+t_1)(q_h-q_2)}$
Sell q_l only	$\frac{[2(q_l-q_2)-k(1+t_1)q_l+k(1+t_2)q_2]^2}{9(1+t_1)(q_l-q_2)}$
Sell both q_h and q_l	$\frac{[2(q_l-q_2)-k(1+t_1)q_l+k(1+t_2)q_2]^2}{9(1+t_1)(q_l-q_2)} + \frac{(1-k(1+t_1))^2}{4(1+t_1)}(q_h-q_l)$
	Case 3 ($q_h > q_2 > q_l$)
Sell q_h only	$\frac{[2(q_h-q_2)-k(1+t_1)q_h+k(1+t_2)q_2]^2}{9(1+t_1)(q_h-q_2)}$
Sell q_l only	$\frac{[q_2-q_l-k(1+t_1)q_l+k(1+t_2)q_2]^2}{9(1+t_1)(q_2-q_l)}$
Sell both q_h and q_l	$\frac{[3(q_h-q_2)+2k(1+t_2)q_2+k(1+t_1)q_2-3k(1+t_1)q_h+\rho]^2}{36(1+t_1)(q_h-q_2)}$ $+\frac{[2k(1+t_2)q_2+k(1+t_1)q_2-3k(1+t_1)q_l+\rho]^2}{36(1+t_1)(q_2-q_l)}$

Table 3: Firm 1's Profits When Demand is Market-wide

3.1 The Least Profitable Strategy

One of Firm 1's strategies is easily ruled out in Case 1 and Case 2. When $q_2 > q_h > q_l$ ($q_h > q_l > q_2$) then exporting both qualities generates more variable profits for Firm 1 than exporting only the high (low) quality. This means that Firm 1 can rule out exporting only that product which is closest in quality to Firm 2's product, since selling both quality levels is unambiguously more profitable. The intuition is clear: if Firm 1's product is close in quality to Firm 2's product, then price competition will be intense, reducing profits. Selling both quality levels is more profitable because the product that is further in quality from Firm 2's product is not subject to such intense price competition and thus generates greater profit margins for Firm 1. Indeed, the product that is further in quality is shielded from competition not only by the greater differentiation in qualities but also by the presence of the product that is closer in quality, which bears the brunt of the competition. Thus Firm 1's least profitable strategy is exporting only the product that is closer in quality to Firm 2's product. Note that this says nothing about whether selling only the product further in quality is more or less profitable than selling both products. It means only that Firm 1 always sells the product that is further in quality, and it may or may not sell the product that is closer in quality.

Proposition 2. *In Case 1, Firm 1's variable profit of selling both qualities is greater than that of selling the high quality only.*

In Case 2, Firm 1's variable profit of selling both qualities is greater than that of selling the low quality only.

Proof. If $q_2 > q_h > q_l$, then $\pi_{h,l} > \pi_h$ iff (see Table 3) $\frac{k^2(1+t_1)}{4}(q_h - q_l) > 0$. Since $q_h > q_l$, this is clearly true.

If $q_h > q_l > q_2$, then $\pi_{h,l} > \pi_l$ iff $\frac{(q_h-q_l)(1-k(1+t_1))^2}{4(1+t_1)} > 0$. Since $q_h > q_l$, then this is clearly true. □

For Cases 1 and 2, then, the decision is in effect between exporting both goods and exporting only the good that is more differentiated from the competitor's good in quality. I refer to these strategies as the two-good strategy and the single-good strategy, respectively.

If on the other hand $q_h > q_2 > q_l$ (Case 3), so that Firm 2's product quality is in between the quality levels that Firm 1 can offer, none of Firm 1's strategies can be ruled out.[25] In Case 1 and Case 2, one of Firm 1's products is by definition always closer in quality to Firm 2's product. This is not true in Case 3: whether the high quality or the low quality is closer to Firm 2's quality depends entirely on the specific parameter values. This means that whether the high quality or low quality good is subject to more intense price competition also varies. In addition, neither good shields the other from competition. And so the argument for Cases 1 and 2 does not apply to Case 3, for which none of the strategies is ruled out.

3.2 Effect of Tariff on the Optimal Strategy

The following lemma concerning Firm 1's market shares is important for understanding the intuition behind Firm 1's optimal strategy.

Lemma 1. *In Case 1 and Case 2, Firm 1's total market share is greater under the two-good strategy than under the single-good strategy if and only if $t_2 > t_1$.*

This result comes directly from comparing the market share under the single-good strategy to the total market share under the two-good strategy as given in Table 4. If Firm 1 has the tariff advantage, then its combined market share under the two-good strategy is always greater. On the other hand, a comparison of prices (see Table 1) shows that prices are always greater under the single-good strategy-regardless of the tariff level of advantage. A tariff advantage means that Firm 1, with prices less inflated by tariffs, is better able to appropriate market share from Firm 2. It does so by introducing a close competitor, a product that is closer in quality to Firm 2's product. If on the other hand Firm 1 is at a tariff *dis*advantage, then offering

[25] This can be seen, for example, by setting parameter values as follows: $k = .5$, $t_1 = .2$, $t_2 = .35$, $q_h = 5$, $q_l = 1$. When $q_2 = 2$, then selling only the high-quality product is more profitable than selling both products. When $q_2 = 3$, then selling only the low-quality product is more profitable.

a close competitor means that Firm 1 loses even more market share to Firm 2 than it would if it offered only the more distant competitor.

	Total demand for Firm 1's products
Firm 1's strategy	Case 1 ($q_2 > q_h > q_l$)
Sell q_l only	$\frac{(q_2-q_l)-k(1+t_1)q_l+k(1+t_2)q_2}{3(q_2-q_l)}$
Sell both q_h and q_l	$\frac{(q_2-q_h)-k(1+t_1)q_h+k(1+t_2)q_2}{3(q_2-q_h)}$
	Case 2 ($q_h > q_l > q_2$)
Sell q_h only	$\frac{2(q_h-q_2)-k(1+t_1)q_h+k(1+t_2)q_2}{3(q_h-q_2)}$
Sell both q_h and q_l	$\frac{2(q_l-q_2)-k(1+t_1)q_l+k(1+t_2)q_2}{3(q_l-q_2)}$
	Case 3 ($q_h > q_2 > q_l$)
Sell q_h only	$\frac{2(q_h-q_2)-k(1+t_1)q_h+k(1+t_2)q_2}{3(q_h-q_2)}$
Sell q_l only	$\frac{(q_2-q_l)-k(1+t_1)q_l+k(1+t_2)q_2}{3(q_2-q_l)}$
Sell both q_h and q_l	$\frac{2(q_h-q_2)(q_2-q_l)+(q_h-q_l)(k(1+t_2)q_2-k(1+t_1)q_2)}{3(q_h-q_2)(q_2-q_l)}$

Table 4: Firm 1's Total Market Share

Determining Firm 1's optimal strategy is the result of comparing the profit equations in Table 3, which produces the following conditions, for Case 1 and Case 2 respectively, under which exporting both products is at least as profitable as the single-good strategy ($\pi_l \leq \pi_{h,l}$ and $\pi_h \leq \pi_{h,l}$):[26]

$$\frac{\left[q_2 - q_l - k(1+t_1)q_l + k(1+t_2)q_2\right]^2}{9(1+t_1)(q_2-q_l)} \leq \frac{\left[q_2 - q_h - k(1+t_1)q_h + k(1+t_2)q_2\right]^2}{9(1+t_1)(q_2-q_h)} + \frac{k^2(1+t_1)}{4}(q_h - q_l) \tag{5}$$

$$\frac{[2(q_h-q_2) - k(1+t_1q_h + k(1+t_2)q_2]^2}{9(1+t_1)(q_h-q_2)} \leq \frac{[2(q_l-q_2) - k(1+t_1q_l + k(1+t_2)q_2]^2}{9(1+t_1)(q_l-q_2)} + \frac{[1-k(1+t_1)]^2}{4(1+t_1)}(q_h - q_l)\,. \tag{6}$$

For ease of exposition, I define the following variables:

[26] Note that the left-hand side is the profit earned from the single-good strategy while the first term on the right-hand side (the profit from selling both goods) is equal to the total profit that Firm 1 would earn by selling only its high quality product (a strategy that has already been ruled out). Note also that the expression in Table 3 for profits from selling both goods consists of two terms, however, it is *not* the case that each term represents profits from only one of the goods.

- ξ is the difference between the left-hand side and the first term on the right-hand side of equation (5) (Case 1);
- $\gamma \equiv \frac{k^2(1+t_1)}{4}(q_h - q_l)$, the second term on the right-hand side of equation (5);
- ζ is the difference between the left-hand side and the first term on the right-hand side of equation (6) (Case 2);
- $\lambda \equiv \frac{\left(1-k(1+t_1)\right)^2}{4(1+t_1)}(q_h - q_l)$, the second term on the right-hand side of equation (6).

Thus, the two-good strategy will be at least as profitable as the single-good strategy if $\xi \leq \gamma$ for Case 1 and if $\zeta \leq \lambda$ for Case 2.

Rearranging, ξ becomes

$$\begin{aligned}\xi =& \frac{(q_h - q_l)}{9(1+t_1)(q_2 - q_h)(q_2 - q_l)} \times \\ & \left[(q_2 - q_h)(q_2 - q_l)\big(1 + k(1+t_1)\big)^2 - (kq_2)^2(t_1 - t_2)^2\right].\end{aligned}$$

Similarly, ζ becomes

$$\begin{aligned}\zeta =& \frac{(q_h - q_l)}{9(1+t_1)(q_h - q_2)(q_l - q_2)} \times \\ & \left[(q_h - q_2)(q_l - q_2)\big(2 - k(1+t_1)\big)^2 - (kq_2)^2(t_1 - t_2)^2\right].\end{aligned}$$

Using these expressions for ξ and ζ, equations (5) and (6) become the following necessary and sufficient conditions, for Case 1 and Case 2 respectively, under which the two-good strategy is no less profitable than the single-good strategy:

$$4\big[kq_2(t_1 - t_2)\big]^2 - (q_2 - q_h)(q_2 - q_l)\left[4 + 8k(1+t_1) - 5k^2(1+t_1)^2\right] \geq 0 \tag{7}$$

and

$$4\big[kq_2(t_1 - t_2)\big]^2 - (q_h - q_2)(q_l - q_2)\left[7 + 2k(1+t_1) - 5k^2(1+t_1)^2\right] \geq 0. \tag{8}$$

These equations are the basis for the following proposition about the effect of tariffs on Firm 1's optimal strategy regarding product proliferation:

Proposition 3. *In Case 1 and Case 2, for any given t_1, if Firm 1's tariff advantage is big enough, then the profit from the two-good strategy is greater than or equal to the profit from the single-good strategy for Firm 1.*

Proof. The left-hand side of equations (7) and (8) is increasing in t_2 if $t_2 > t_1$. If t_2 is big enough to give Firm 1 the tariff advantage, then a big enough increase in t_2 will cause the left-hand side of the equation to become positive. Therefore if t_2 is big enough the equations will hold. □

The next proposition addresses the effects of a fall in t_1 on Firm 1's profits.[27]

Proposition 4. *In Case 1 (Case 2), if Firm 1's tariff is small enough and $k < \frac{4}{5}$ ($k < \frac{1}{5}$) then the left-hand side of equation* (7) *(*(8)*) increases as Firm 1's tariff falls.*

Proof. Case 1: Taking the partial derivative of (7) with respect to t_1 gives $8k^2q_2^2(t_1 - t_2) - (q_2 - q_h)(q_2 - q_l)[8k - 10k^2(1 + t_1)]$. As $t_1 \to 0$, this evaluates to $-8k^2q_2^2t_2 - (q_2 - q_h)(q_2 - q_l)(8k - 10k^2) < 0$ (strictly) for any value of t_2 when $k < \frac{4}{5}$. Since the derivative is continuous in t_1, it will be negative when t_1 is small enough. The proof for Case 2 is analogous. □

If the conditions of both Proposition 3 and Proposition 4 are met, then product proliferation becomes profitable when Firm 1's tariff falls far enough.

Corollary 1. *In Case 1 (Case 2), if Firm 1 has a small enough tariff and a big enough tariff advantage and $k < \frac{4}{5}$ ($k < \frac{1}{5}$), then a further reduction in in t_1 will result in the two-good strategy being more profitable.*

Intuitively, two effects combine to produce the above propositions. First, exporting a second product provides Firm 1 with a closer competitor that can take more market share away from Firm 2. The additional product that Firm 1 chooses to export is the one that is closer in quality to Firm 2's product. Since it competes more directly with Firm 2's product, it is better able to acquire market share. But this effect is only present when Firm 1 has the tariff advantage, since only in this case will its price be competitive enough to take market share away from Firm 2. In addition, this strategy is profitable only if the increase in market share is sufficient to override the additional intra-firm competition introduced by seeling two products, which will be the case only when Firm 1's tariff is low enough. If Firm 1's tariff is too high, then it must set a low price to attract enough customers (regardless of whether it adopts the single-good or the two-good strategy). This effect is compounded by the intra-firm competition introduced by the two-good strategy. However, as Firm 1's tariff falls, it can set a higher price, which due to the tariff advantage still allows it to compete effectively with Firm 2 to acquire sufficient market share while also maintaining sufficiently high per-unit profits (even in the face of intra-firm competition). Product proliferation, which entails the introduction of a buffer product by Firm 1, is therefore only profitable for Firm 1 when tariffs are low enough.

[27] If $t_1 \times t_2 < \frac{9}{4}$ and the restrictions on k in Proposition 4 apply, then Proposition 3 necessarily implies Proposition 4.

3.3 Effect of Quality on the Optimal Strategy

As noted above, profits from the two-good strategy will be greater than profits from the single-good strategy when

- in Case 1, $\xi < \gamma$
- in Case 2, $\zeta < \lambda$.

The next proposition provides additional conditions under which the two-good strategy is the optimal choice.

Proposition 5. *In Case 1 and Case 2, if the difference in quality between Firm 1's products and Firm 2's product is small enough and Firm 1 has the tariff advantage, then the profit from the two-good strategy is greater than or equal to the profit from the single-good strategy for Firm 1.*

Proof. For Case 1, equation (7) gives the condition under which the two-good strategy is no less profitable than the single-good strategy. The first term is *strictly* positive when Firm 1 has the tariff advantage. If $k(1+t_1) < 2$, then the second term is decreasing in q_h and q_l. Since the second term is continuous in q_h and q_l and approaches 0 as $q_h, q_l \to q_2$, then for any q_2 there is some q_h and q_l small enough such that the second term is smaller than the first term and the expression holds. If $k(1+t_1) < 2$, then the second term is negative, and the whole expression holds regardless of quality difference.

The proof for Case 2 proceeds analogously. □

The intuition is developed using Case 2 as an example. If the quality differences ($q_h - q_2$ and $q_l - q_2$) are small enough, then Firm 1 can benefit from what I refer to as an "aggressor product". With a large quality difference, price competition is not very intense. In this case, Firm 1 is better off selling only its high-quality product. In the presence of minimal price competition, Firm 1 can set a high price, thereby earning a high per-unit profit. At the same time, Firm 2 also sets a relatively high price so that Firm 1 does not lose too much market share due to its own high price. Thus, with a large quality difference, limited price competition allows Firm 1 to earn a high profit margin per unit while maintaining sufficient market share with only its high-quality product.

However, if qualities are close and price competition is intense, then Firm 1 would lose significant market share by selling only its high-quality product and setting a high price (which generates a high per-unit profit). Instead, Firm 1 chooses to pursue market share more aggressively by exporting both qualities, using the low-quality good as an "aggressor product". This good competes intensely with Firm 2's product, capturing more market share from Firm 2 than the high-quality product alone could. When qualities are

close, price competition is intense and Firm 1 is no longer able to set such a high price per unit. Firm 1 therefore switches its strategy to focus on market share instead.

To summarize, when quality differences are small, price competition is severe. In this instance, Firm 1 benefits from having an "aggressor product," which competes intensely with Firm 2 for market share. Even though the presence of this product creates intra-firm competition which lowers prices, due to the intensity of competition, Firm 1 is better off competing for market share when qualities are close and therefore uses the aggressor product to challenge Firm 2 for market share. This combination produces greater profits under the two-good strategy than the single-good strategy.

3.4 Case 3: $q_h > q_2 > q_l$

In Case 3, there is no single "aggressor product;" Firm 1's high-quality product competes with Firm 2's product from above, and Firm 1's low-quality product competes with Firm 2's product from below. Also, in Case 3, Firm 1's products never compete with each other directly; any competitive effect one product has on the other is only through its effect on Firm 2's product. This means that Firm 1's products never internalize their competitive effects on each other, a critical feature of Case 3.

As seen in Section 3.1, when Firm 2 offers the middle quality, none of Firm 1's strategies can be ruled out. However, it is true that per-unit profits are never greatest when Firm 1 exports both goods in Case 3.

Proposition 6. *If $\frac{q_2-q_l}{q_h-q_2} < \frac{2-k(1+t_1)}{1+k(1+t_1)}$, then Firm 1 earns its highest per-unit profit $(p_h - c_h)$ in Case 3 when exporting only its high-quality good. If $\frac{q_2-q_l}{q_h-q_2} > \frac{2-k(1+t_1)}{1+k(1+t_1)}$, then Firm 1 earns its highest per-unit profit $(p_l - c_l)$ in Case 3 when exporting only its low-quality good.*

Proof. From Table 5 it is easy to see that the profit per unit for the high-quality good is always greater when selling only the high-quality good than when selling both goods. Similarly, the profit per unit for the low-quality good is always greater when selling only the low-quality good than when selling both goods. Under the two possible single-good strategies, the per unit profit for the high-quality good is greater than the per-unit profit for the low-quality good if and only if

$$\frac{2(q_h - q_2) + k(1+t_2)q_2 - k(1+t_1)q_h}{3(1+t_1)} > \frac{(q_2 - q_l) + k(1+t_2)q_2 - k(1+t_1)q_l}{3(1+t_1)}.$$

This simplifies to $\frac{q_2-q_l}{q_h-q_2} < \frac{2-k(1+t_1)}{1+k(1+t_1)}$.[28] When this expression is true, then the profit per unit when selling only the high-quality good is greater than the profit per unit when selling only the low-quality good. But the profit per unit of selling only the low quality is always greater than the profit per unit on the low-quality

[28] If $k(1+t_1) > 2$, this is never true, since the right-hand side will be non-positive and the left-hand side will be positive.

good when selling both goods. By transitivity, the profit per unit when selling only the high quality good must therefore be greater than the profit per unit on the low-quality good when selling both goods. And the profit per unit of selling only the high quality is always greater than the profit per unit on the high-quality good when selling both goods. Therefore, under this condition, the greatest profit per unit is earned by selling only the high quality good. A *mutis mutandis* argument for the opposite condition completes the proof. □

	Firm 1's per-unit profit
Firm 1's strategy	Case 1 ($q_2 > q_h > q_l$)
Sell q_l only	$\frac{(q_2-q_l)-k(1+t_1)q_l+k(1+t_2)q_2}{3(1+t_1)}$
Sell both, q_l per-unit profit	$\frac{2(q_2-q_h)-3k(1+t_1)q_l+k(1+t_1)q_h+2k(1+t_2)q_2}{6(1+t_1)}$
Sell both, q_h per-unit profit	$\frac{q_h-q_2+k(1+t_2)q_2-k(1+t_1)q_h}{3(1+t_1)}$
	Case 2 ($q_h > q_l > q_2$)
Sell q_h only	$\frac{2(q_h-q_2)-k(1+t_1)q_h+k(1+t_2)q_2}{3(1+t_1)}$
Sell both, q_h per-unit profit	$\frac{3(q_h-q_2)+(q_l-q_2)-3k(1+t_1)q_h+2k(1+t_2)q_2+k(1+t_1)q_l}{6(1+t_1)}$
Sell both, q_l per-unit profit	$\frac{2(q_l-q_2)+k(1+t_2)q_2-k(1+t_1)q_l}{3(1+t_1)}$
	Case 3 ($q_h > q_2 > q_l$)
Sell q_h only	$\frac{2(q_h-q_2)-k(1+t_1)q_h+k(1+t_2)q_2}{3(1+t_1)}$
Sell q_l only	$\frac{(q_2-q_l)-k(1+t_1)q_l+k(1+t_2)q_2}{3(1+t_1)}$
Sell both, q_h per-unit profit	$\frac{3(q_h-q_2)-3k(1+t_1)q_h+2k(1+t_2)q_2+k(1+t_1)q_2+\frac{(q_h-q_2)(q_2-q_l)}{q_h-q_l}}{6(1+t_1)}$
Sell both, q_l per-unit profit	$\frac{2k(1+t_2)q_2+k(1+t_1)q_2-3k(1+t_1)q_l+\frac{(q_h-q_2)(q_2-q_l)}{q_h-q_l}}{6(1+t_1)}$

Table 5: Firm 1's Profit Margins

Intuitively, if the quality difference between q_h and q_2 is relatively large compared to the difference between q_2 and q_l, then competition between the first pair is less intense than between the second pair. Firm 1 is free to set a high price for its high-quality product and consequently to earn a large profit per unit. Since competition between the low-quality good and Firm 2's product is more intense, Firm 1 is forced to lower its price and earn a smaller profit per unit on the low-quality good.

Profits per unit on the high-quality will also be lower when selling both goods because Firm 1 does not fully internalize the competition that it creates for itself by selling both goods. If Firm 1 sells both of its goods, then the low-quality good competes with Firm 2's good. This price competition causes Firm 2 to set a low price, which in turn forces Firm 1 to lower the price on its own high-quality good in order to maintain sufficient market share for that good. This lowers the profit per unit when compared to selling only the high-quality good. Thus when the difference between q_h and q_2 is large, the greatest per-unit profit is when Firm 1 sells only its high-quality good. Similar reasoning explains why selling only the low-quality good generates the greatest profits when the difference between q_2 and q_l is larger.

The role of the tariff in this relationship is also intuitive: as Firm 1's tariff increases, it lowers prices. If quality differences are symmetric, then this effect is greater for the high-quality product.[29] This is because the tariff has a multiplicative effect on price, so in order to keep the price seen by consumers ($(1+t_1)p_h$) low enough to retain sufficient market share, Firm 1 must lower the price of its high-quality good more than it lowers the price of its low-quality good when its tariff increases. Thus, with a high tariff, the per-unit profit of the high-quality good is likely to be less than the per-unit profit of the low-quality good.

According to Proposition 6, the greatest profit per unit is generated under one of the two single-good strategies. Then the only situation in Case 3 in which Firm 1 will choose to export both products is one where this strategy allows Firm 1 to capture enough market share to make up for the lower per-unit profits. The necessary (but not sufficient) conditions for the two-good strategy to deliver the greatest profits are therefore:

- If selling only the low-quality good delivers the highest profit per unit, the combined market share under the two-good strategy is greater than the market share under the low-quality only strategy, and
- If selling only the high-quality good delivers the highest profit per unit, the combined market share under the two-good strategy is greater than the market share under the high-quality only strategy.

Comparing the market shares in Table 4 produces the following proposition:

Proposition 7. *Firm 1's combined market share under its two good strategy is greater than its market share under the high-quality only strategy and the low-quality only strategy, respectively, when*

$$k(1+t_2)q_2 > k(1+t_1)q_l$$

$$q_h - q_2 + k(1+t_2)q_2 > k(1+t_1)q_h.$$

Clearly, if $t_1 < t_2$, then the first expression will necessarily be satisfied and Firm 1's market share under the two-good strategy is bigger than the market share under the high-quality only strategy in Case 3.[30] The second expression will necessarily be satisfied if Firm 1 has the tariff advantage and $k(1+t_1) < 1$.[31]

This relation between the market shares under each strategy supplies the intuition for comparing profits under each strategy. Suppose, for example, that selling only the low-quality good generates the greatest profit per unit for Firm 1. Then Firm 1's market share under the two-good strategy must be greater than its market share under the low-quality strategy in order for the two-good strategy to be more profitable. This is possible only if, by also introducing its high-quality good to the market, Firm 1 is able to steal a significant

[29] To be precise, if $2(q_h - q_2) > q_2 - q_l$, then this effect is greater for the high-quality product.

[30] Having the tariff advantage is sufficient but not necessary; it is possible for the market share under the two-good strategy to be greater even if Firm 1 does not have the tariff advantage, as long as the quality difference is big enough.

[31] Again, this is sufficient but not necessary.

portion of Firm 2's high-valuation customers (those customers with high θ values). This will tend to happen the less differentiated in quality are Firm 1's high-quality product and Firm 2's product, as this will increase competition and make the high-quality product more attractive to consumers (since it is higher in quality and, due to intense competition, relatively low in price).[32] Thus the results here are similar to the results in Cases 1 and 2: when Firm 1 has the tariff advantage, then if the quality difference between Firm 1 and Firm 2 is small enough, the two-good strategy will be more profitable (in Cases 1 and 2) or capture a greater market share (in Case 3). In Cases 1 and 2, a small quality difference is a sufficient condition for Firm 1 to choose product proliferation; in Case 3, it is necessary in order for the market share under proliferation to be greater, but not sufficient to cause profits under proliferation to be greater.

4 The Automobile Market in Mexico

The automobile market in Mexico is a good environment for testing the predictions about quality proliferation in export markets. Having liberalized auto import policies with the 1989 Mexican Automotive Decree that reversed a 1962 ban, Mexico has imported significant numbers of automobiles since the early 1990s (Hufbauer and Schott, 2005). Sales of new cars (exclusive of light trucks) in Mexico were just 271,260 in 1989 but more than doubled to 585,618 by 2008; many of these cars were imported.[33] Even though some popular models are produced locally, including the VW Jetta, the Chevrolet Chevy, and the Nissan Tsuru, a majority of the car models sold in Mexico are still imported.

From 1989 when Mexico imported only one car model (the Oldsmobile Cutlass), the choices available to new car buyers in Mexico have increased manyfold. In 1990, only five auto manufacturers were actively selling vehicles in Mexico: Chrysler, Ford, General Motors, Nissan, and Volkswagen. These manufacturers sold a total of 12 different makes (brands) of car, with Chrysler selling Chrysler, Dodge, and Plymouth cars, Ford selling Ford, Lincoln, and Mercury cars, and GM seling Buick, Cadillac, Chevrolet, and Oldsmobile. Nissan and Volkswagen sold just one make each. By 1997, the number of firms active in the market had increased to nine with the addition of BMW, Mercedes-Benz, Honda, and Porsche.[34] And by 2009, Acrua, Audi, Fiat, Jaguar, Mazda, Mitsubishi, Peugeot, Renault, Saab, Subaru, Toyota, SEAT, and Volvo had joined the competition for new car buyers, bringing the number of manufacturers to 22.[35] The number of car brands also increased, from 12 in 1990 to 18 in 1997 and 30 in 2009. Table 6 shows the details of entry

[32] Even if $t_2 < t_1$, then it is possible that Firm 1's combined market share will be greatest under the two-good strategy than under either single-good strategy. This would require a very large quality differentiation between Firm 1's products and Firm 2's product.

[33] Car production in Mexico grew even faster than car sales. In 1989, Mexico produced 435,250 cars (exclusive of light trucks). This figure grew to 1,312,024 by 2007. A large portion of these cars were made for export.

[34] The number of Porsche cars sold was very small (fewer than 100) and remains small in 2009. If light trucks are included, the number of firms active in Mexico in 1997 was 10, as a small number of Land Rovers were also sold by that time.

[35] Three car makers, Alfa Romea, MG, and Infiniti, both entered and exited the Mexican market between 1990 and 2009.

Year	Number of Car Brands	New Entrants
1997	11	
1999	13	Jaguar, Peugeot
2000	14	Volvo
2001	17	Renault, Saab, Seat
2002	20	Cadillac, Mini, Toyota
2003	23	MG Rover, Mitsubishi, Pontiac
2004	23	Smart; Pontiac (exit)
2005	24	Acura
2006	31	Bentley, Fiat, Mazda, Mercury, Pontiac, Subaru, Suzuki
2007	31	Alfa Romeo; MG Rover (exit)
2008	35	Faw, Ferrari, Maserati, Maybach

Table 6: Car Brands Available in Mexico

Year	Compact	Midsize	Luxury	Sports
1997	5	23	21	11
1999	9	28	32	10
2000	10	31	36	9
2001	16	32	37	10
2002	17	33	41	13
2003	22	34	45	18
2004	22	31	47	17
2005	25	30	53	17
2006	30	37	58	18
2007	34	34	58	19
2008	38	37	68	19

Table 7: Mexico: Number of Models per Year by Segment

into the new car market from 1997 to 2008.

Not only did many new firms enter the Mexican auto market, most of them introduced multiple car models. Table 7 gives the number of models available for sale in each segment of the Mexican auto market. While the choices for consumers increased in each of the four segments, those in the compact and luxury segments multiplied especially rapidly.

Although it has grown significantly, the Mexican auto market is still a relatively small piece of the global market. In 2008, according to Standard & Poor's/Bloomberg Businessweek, global light vehicle sales totaled 66.785 million. Light vehicle sales in Mexico in the same year were just over 1 million, accounting for 1.5% of world sales. Changing conditions in the Mexican market, including changes in trade costs, are therefore unlikely to induce auto manufacturers to substantially alter their model design.

Both the automobile industry and the new car market in Mexico have been impacted by a number of trade agreements that were negotiated in the two decades since the 1989 Decree: NAFTA in 1994, the EC-Mexico

Free Trade Agreement in 2000, and the Japan-Mexico Free Trade Agreement in 2005 are particularly critical for auto imports. In addition, by 2010, Mexico also had trade agreements with Chile, Colombia, Costa Rica, El Salvador, Guatemala, Honduras, Nicaragua, and Israel. Mexico is also a member of the Global System of Trade Preferences among Developing Countries (GSTP) and the Latin American Integration Association (ALADI), the latter of which has included significant reciprocal tariff concessions on automobiles. On the other hand, some auto exporters such as China and South Korea have not benefitted from any preferential tariff liberalization. As the Mexican auto market since 1989 has been characterized by both significant proliferation in the number and quality of products as well as significant variance in tariffs, it provides a good test case for empirically evaluating the relationship between quality proliferation on the one hand and trade costs and quality differences on the other.

4.1 Data

The Mexican Automobile Dealer Association (Asociación Mexicana de Distribuidores de Automotores A.C., AMDA) compiles raw monthly data on the prices of the makes, model, and trim level or option package of new car for sale at dealerships throughout Mexico.[36] These data also assign each car to one of the following categories: subcompact, compact, luxury, or sports. The make of a car is its manufacturer (e.g. Toyota, Ford, Renault, etc.) while the model is the type of car (e.g. Camry, Focus, Megane, etc.). Each car model typically is available in multiple trim levels or packages, which I label as versions for ease of reference throughout the remainder of the paper. I used the data for 1997 through 2008, with the exception of 1998 (for which the data were not available).

Car models from a single manufacturer typically differ in physical dimensions (for example, the Honda Civic and Accord), while versions do not.[37] The versions of a particular model typically differ in terms of upgrades, such as air conditioning, leather seats, or high-end sound systems, and horsepower.[38]

I identify each car model as a separate product and each version as a separate quality level. The overall quality of a car derives from multiple attributes. McCarthy (1996) and Wojcik (2000), for example, find that physical dimensions, horsepower (or horsepower per pound), cost per mile driven, climate control, market segment, manufacturer, and safety rating all impact the demand for a model. That is, these characteristics affect consumers' willingness to pay, as they determine the (perceived) quality of the car. A given car model generally has the same safety rating and physical dimensions across trim levels/option packages and engines.[39] Clearly each car model is also made by the same manufacturer. I therefore use horsepower and

[36] The models in the AMDA data are consistent with data from Ward's Automotive, which lists sales at the make-model level but not the make-model-trim level.

[37] In some cases, a single car model may be available in different body styles, such as coupe and 4-door.

[38] A few models in Mexico are even available with a "security package."

[39] The exception is when a single model is sold in multiple body styles, such as coupe and 4-door.

trim levels/option packages, which differ in exactly the remaining quality attributes that affect willingness to pay, to identify the different quality levels of each model that are sold in the Mexican market each year.

Using the AMDA data supplemented by specifications for each model, I construct two different count variables of the number of versions. The first count variable, labeled Count1, counts each distinct combination of trim level/option package, horsepower, body style, and transmission as a separate quality level. This count variable uses both the attributes that differ across trim levels as well as any differences in physical dimensions, as suggested by McCarth (1996) and Wojcik (2000), to identify quality levels. The second count variable, labeled Count2, counts only distinct combinations of trim level/option package, horsepower, and body style but not transmission, since transmission type has not previously been identified as a component of a car's quality.[40]

The data on tariffs are gathered from the World Integrated Trade Solution (WITS), which incorporates data from the UN Comtrade database (from the United Nations Statistical Division), the TRAINS database (from the United Nations Conference on Trade and Development), and the IBD and CTS databases (from the World Trade Organization).[41] Since Mexican auto tariffs are based on engine displacement, tariff rates are assigned to each observation based on the year of sale, country of assembly, and engine size. For example, the tariff rate in 2001 on any car imported from Canada was 4.4%, while the tariff rate in 2003 on cars with engine displacements between 1001 and 1500 cc from Japan was 30%. As noted in Section 4, there is significant variation in tariff rates across countries and years due to the signing and phasing in of multiple free trade agreements. The tariff rates are the applied rates for each year (as opposed to the bound rates).

Using data on the specifications for each car, I also calculate a series of variables to represent the degree of quality differentiation between a product and its competitors. Specifically, I use data on physical dimensions, horsepower, engine size, fuel efficiency, and cost per mile driven to calculate these variables.[42] The quality difference variable is calculated as the absolute value of the difference between the quality measure of a given model and the average of the same measure for all other cars in the same segment in the same year.[43] This calculation of quality differences therefore also accounts for the segment in which a car competes–another important quality attribute. Data on the safety rating of cars were unfortunately not available and I therefore

[40]Model years in Mexico closely track those in the United States, and therefore multiple model-years were often sold in any given calendar year. If two model-years of an otherwise identical version were sold in the same calendar year, then these were counted as a single version of the car.

[41]See Data Appendix for complete list of data sources.

[42]As noted above, these attributes are commonly used to represent the quality of a vehicle. While climate control is also clearly a determinant of quality, data for this attribute was not consistently available. Furthermore, many of the car models within a market segment were available in versions both with and without climate control, and therefore there is very little variation in this quality measure between models within a segment.

[43]While physical dimensions do not often vary across versions of a single model-year, other attributes such as horsepower and fuel efficiency do. The quality level (with respect to these attributes) is thus not always constant across a single model. To account for this within-model variation, I calculate the degree of quality difference using the median, mean, minimum, and maximum of these attributes for each model. That is, I calculate the degree of quality difference using four different ways of determining the quality level of each model. As expected, there is a very high correlation between the quality difference variables calculated using these various quality measures per model.

cannot calculate a quality difference measure for this attribute. However, since safety ratings tend to be consistent for cars produced by a single manufacturer, including a manufacturer dummy variable will control for much of this difference.

There are a total of 1,235 observations at the year-make-model level, for the years 1997 and 1999 through 2008. For 26 of these observations (3 car models), complete data on specifications (quality proxies) is not available; these are not included in the summary data or the empirical estimates.[44] Summary statistics are provided in Table 8.

Variable	Obs	Mean	Std. Dev.	Min	Max
Count Variable 1	1209	5.86	6.11	1	64
Count Variable 2	1209	2.95	2.22	1	15
Ad valorem tariff	1209	.106	.140	0	.5
Car Size ($\frac{l \times w \times h}{1000}$)	1209	698.5	107.6	355.4	1125
Wheelbase	1209	103.9	7.57	71.2	142.1
Horsepower	1209	199	98.2	51	626
Fuel efficiency	1209	23.3	5.42	10.7	42.7
Engine size	1209	2.79	1.25	.6	8.4
Price (thousands of pesos, base year 2005)	1209	498762.5	558846.2	71571.2	6104374
Difference between quality measure and average for segment					
Wheelbase	1209	5.4	4.3	.015	33.2
Horsepower	1209	49.9	52.0	.027	335.3
Fuel efficiency	1209	3.05	2.81	.004	20.08
Engine size	1209	.747	.705	0	5.15

Table 8: Summary Statistics

5 Empirical Analysis of Product Proliferation

The focus of the theoretical model is on the effect of tariffs on product quality proliferation The analytical results of section 3 provide two testable hypotheses. The first is that product quality proliferation is greater the lower is the tariff. The second is that product quality proliferation is greater the smaller is the degree of quality differentiation.

5.1 Empirical Specification

To examine these hypotheses empirically, I regress the count of trims levels or packages of each model-year on the tariff level and the degree of quality differentiation. Both variables should have a negative effect on the number of quality levels available. In order to control for variety and the growth of market size, I also

[44] Both qualitative and quantitative results were similar when these observations were included in the analysis.

include brand and year dummies. The estimating equation is

$$log\left(E[Count_{i,y}]\right) = \beta_0 + \beta_1 t_{j,y} + \beta_2 \Delta q_{i,y} + \bar{\gamma}\bar{y} + \bar{\kappa}\bar{b} \tag{9}$$

where $Count_{i,y}$ is the number of trim levels offered in year y of make/model i, $t_{j,y}$ is the tariff rate applied to auto imports from country j (the country of assembly of make/model i) in year y, $\Delta q_{i,y}$ is the degree of quality differentiation, $\bar{y}$ is a series of dummy variables (fixed effects) for the year, and $\bar{b}$ is a series of dummy variables (fixed effects) for the car brand (make). The brand-level fixed effects will control for the styling of the car, as this is in general constant across years as well as the various models of a brand.[45]

Because the right-hand side variable is discrete, there is a choice between using the Poisson and negative binomial distributions of the count variable in estimating the β coefficients. Furthermore, because I only observe cases where at least one trim level of each model is sold, truncated distributions are the appropriate distributional assumption. Finally, there is likely to be some correlation between error terms. For example, unobserved variables may include sales and marketing strategies at the model level or corporate events at the brand level. Both of these will likely lead to some serial correlation and within-brand correlation of the unobserved variables. Some correction of the standard error estimates is therefore necessary. Table 14 in the Data Appendix shows the results of simple estimations with a single explanatory variable using both distributional assumptions and three different calculations of standard errors.[46]

As Figure 2 shows, the distribution of the count variable is closer to a binomial distribution than a Poisson distribution. This conclusion is further supported by the fact that the variance of the count variable is significantly different from the mean and that the estimates of α in Table 14 are significantly different from zero, both of which are indicators of overdispersion. The better distributional assumption is therefore a truncated binomial distribution. Table 14 also shows that standard error estimates are most conservative when clustered at the brand level.[47] I therefore use a truncated negative binomial distribution maximum likelihood estimation with standard errors clustered at the brand level for the empirical analysis.

5.2 Baseline Estimation Results

The results of the baseline regression, with the tariff rate and the difference from the sector average quality as explanatory variables, are given in Table 9. The table shows the results when each of the following three characteristics is used as a proxy for quality: engine size (displacement), physical size of the car, and dollars

[45] For example, BMW styling is distinct from that of Audi; Porsche styling is distinct from Jaguar; Renault styling is distinct from Peugeot.

[46] Results using the second count variable are qualitatively and quantitatively similar, although the estimated coefficient on the tariff is slightly smaller in magnitude.

[47] There are 36 clusters; see Cameron, Gelbach, and Miller (2008) for a discussion of how many clusters is too few.

per mile of operating the car.[48] Results are given when each quality proxy is included individually and when all three are included.

	Model 1	Model 2	Model 3	Combined Model
Tariff Rate	-1.18*	-1.18*	-1.21*	-1.20*
	(.524)	(.537)	(.524)	(.529)
Engine Size	-.229**			-.192**
	(.069)			(.073)
Cost per Mile		-7.55		-1.55
		(4.40)		(4.70)
Physical Size			-.000	-.000
			$(1.02 \times e^{-6})$	$(1.22 \times e^{-6})$
Year FE	yes	yes	yes	yes
Brand FE	yes	yes	yes	yes
Constant	-17.7**	-18.4**	-18.2**	-17.8**
	(.949)	(.850)	(1.05)	(1.07)
Observations	1209	1209	1209	1209
ln(L)	-2952.6	-2960.6	-2962.0	-2950.2
** Significant at the 1% level; * significant at the 5% level.				

Table 9: Effect of Tariff and Quality Difference on Number of Versions

The sign on the tariff variable is negative and statistically significant in all four regressions, indicating that a higher applied tariff results in a lower number of versions of a car model being sold in the Mexican market. Estimates from the first model, which uses engine size as the quality proxy, predict that when the tariff falls from .2 to 0 (with all other variables held constant), the expected number of versions will rise from 4.9 to 6.3, an increase of more than 1. The coefficient estimate is therefore also economically significant. This is consistent with the analysis in Section 3: quality proliferation is not profitable when trade costs, in this case tariffs, are too high. Note that the estimated coefficient on the tariff variable remains consistent between all four different models, suggesting that the result is quite robust.

Each of the proxies for the magnitude of the quality gap have the expected negative sign. That is, a larger quality difference results in the sale of fewer versions in the Mexican market. Only the engine size, however, is statistically significant. This suggests that the relative power of a car, which is largely due to engine displacement volume, has an effect on the proliferation of versions of the product.[49] If the power that a car delivers is similar to that of other cars in its segment, then the firm decides to sell more versions in the market. Because quality levels are close, selling more versions enables a firm to steal so much of the competitor's market share that this outweighs the fact that the firm is also thereby driving its own

[48] The three other characteristics of each car are highly correlated with these three characteristics (see Table 15 in Data Appendix), and therefore the results using these alternative measure of quality are not included here. Note that the correlation between fuel efficiency and the other characteristics is negative because better fuel efficiency typically results from a smaller engine and a smaller car, and better fuel efficiency translates into a lower cost per mile of operation. Results using these alternative quality measures are qualitatively similar.

[49] While torque is actually the primary determinant of a car's pickup, horsepower and engine size are more frequently referenced in advertising and are therefore critical to perceptions of engine power.

price lower through more intense competition. The brand-level fixed effects, not reported here, were almost all significant, indicating that brand-level characteristics, such as demand for car styling, also impact the proliferation decision.

In the combined model, the magnitude of the coefficient on both engine size and cost per mile is reduced. This is likely because engine size has a significant impact on cost per mile: more displacement in the engine tends to burn more fuel. Indeed, the correlation between these two quality measures is .71, while the correlation between the quality difference measure using each of these characteristics is .56. The significant amount of collinearity likely impacts the coefficient estimates.

It is perhaps surprising that the degree of difference in the cost per mile of driving does not have a significant effect on the number of versions. There are a number of possible explanations for this. First, Mexican car buyers may simply be less price conscious with respect to the cost per mile driven and therefore may not consider this as a quality indicator. Second, it is possible that only some market segments include cost per mile in their evaluation of quality. Third, it is possible that Mexican consumers do consider cost per mile to be relevant but that the quality gap is not a significant factor in quality proliferation; in this case, there is some other factor driving the significance of the quality difference measure for engine size.

5.3 Alternative Measures of Tariffs and Quality Difference

Propositions 3 and 5 require not just that Firm 1 have a low tariff but in particular that Firm 1 have a low tariff *advantage* in order for proliferation of qualities to be profitable. Thus, I also estimate equation (9) using the tariff advantage instead of the tariff level as the regressor. The tariff advantage is calculated as the difference between the trade-weighted tariff (for a given year and tariff line) and the applied tariff for the country of assembly. As Table 10 shows, the results regarding the tariff are consistent with both the theoretical results as well as the empirical analysis in Section 5.2. That is, a bigger tariff advantage leads to quality proliferation by increasing the number of model versions available. In particular, when the tariff advantage increases from 0 to .1, then (with all other variables held at their averages), the number of versions increases from 5.6 to 6.3. Using the tariff advantage instead of the tariff level also impacts the estimates of the effect of quality differentiation. In particular, the coefficient on the cost per mile variable is both statistically and economically significant.

To account for the possibility that some market segments may value different quality measures more or less, I add interaction terms to the estimation. Specifically, I interact a dummy variable for cars in the high-end luxury and sports segments with each of the three quality difference measures. The results are presented in Table 11.

	Model 1	Model 2	Model 3
Tariff Rate	-1.18*	-1.18*	-1.21*
	(.524)	(.537)	(.524)
Engine Size	-.229**		
	(.069)		
Cost per Mile		-7.55	
		(4.40)	
Physical Size			-.000
			$(1.02 \times e^{-6})$
Year FE	yes	yes	yes
Brand FE	yes	yes	yes
Constant	-17.7**	-18.4**	-18.2**
	(.949)	(.850)	(1.05)
Observations	1209	1209	1209
ln(L)	-2952.6	-2960.6	-2962.0
** Significant at the 1% level; * significant at the 5% level.			

Table 10: Effect of Tariff Advantage and Quality Difference on Number of Versions

For all three quality difference measures, the dummy variable for car that competes in the high-end segments is negative and significant: models in the high-end segments tend to have fewer versions for sale in the market. This may be explained by the fact that even the base versions of high-end cars typically come with climate control systems and leather seats, so that there may simply be fewer variations left in terms of options and upgrades.

The interaction terms are all positive, and two of them are significant at the 5% level. For these same two variables, cost per mile and physical size, including the interaction terms also makes the quality difference measure itself significant. In other words, quality differences, as measured in terms of these two variables, have a significantly negative impact on product proliferation in the passenger car market but not in the luxury and sports car markets. Recall the intuition that when quality differences are small, competition is intense regardless of how many versions (quality levels) a firm sells. In this case, quality proliferation will cause the firm to gain more in market share from its competitor than it loses due to the lower prices that results from the increased competition. Quality differences may be less important in high-end markets simply because consumers have a strong preference for a certain make or model of car (e.g. a consumers may strongly prefer a Porsche over a Jaguar). In addition, most versions in high-end markets are likely to be of high quality. Thus, quality may not be enough to override make or model preferences for consumers. In this case, competition in quality may be very low, regardless of the magnitude of quality differences.

	Model 1	Model 2	Model 3
Tariff Rate	-1.08*	-1.07*	-1.05*
	(.441)	(.431)	(.470)
Market Segment	-.739**	-.842**	-.864**
(1=luxury or sports)	(.251)	(.192)	(.175)
Engine Size	-.316**		
	(.109)		
Engine Size × high-end	.286		
	(.179)		
Cost per Mile		-18.5**	
		(5.57)	
Cost per Mile × high-end		17.4*	
		(7.81)	
Physical Size			$(-3.66 \times e^{-6})$**
			$(1.25 \times e^{-6})$
Physical Size × high-end			$(3.64 \times e^{-6})$*
			$(1.64 \times e^{-6})$
Year FE	yes	yes	yes
Brand FE	yes	yes	yes
Constant	-16.8**	-16.8**	-16.6**
	(1.03)	(1.35)	(1.07)
Observations	1209	1209	1209
ln(L)	-2921.1	-2921.1	-2919.7
** Significant at the 1% level; * significant at the 5% level.			

Table 11: Estimating Effects by Market Segment

5.4 Robustness

While year fixed effects account for some variation across years, it is also possible that an increase in the number of versions sold is simply the result of greater market experience. As the firm gains experience and finds success in the market, it adopts a product proliferation strategy. This may happen, for example, as the firm trains salespeople, parts distributors, and repair technicians. To control for this, I drop the first year of entry of a model into the market and also use a time trend. The results are given in Table 12. While the coefficient on the tariff variable is smaller with a time trend as expected, it remains significant both economically and statistically. The results are similar when dropping the outliers: approximately 5% of observations have more than 20 versions for sale in a given year. Of these one BMW model (the 3 series) and one Audi model (the A3) have more than 30 counts in some years.[50] Dropping these observations from the estimation reduces the magnitude of the coefficient on tariffs and on the quality difference (engine size) variables slightly, but the relationship remains the same. It therefore appears that the results are not being driven by these outliers.

[50] Dropping all observations of these two models marginally increases the magnitude of the coefficient on tariff and increases its significance to the 1% level.

	Exclude Year 1	Time Trend	Versions ≤ 20
Tariff Rate	-1.25*	-0.96*	-1.05*
	(.523)	(.453)	(.503)
Engine Size	-.250**	-.252**	-.214**
	(.074)	(.074)	(.053)
Time Trend		.042*	
		(.017)	
Year FE	yes	no	yes
Brand FE	yes	yes	yes
Constant	-17.7**	-18.0**	-18.1**
	(.904)	(1.92)	(.355)
Observations	1043	1043	1160
ln(L)	-2613.9	-2619.9	-2652.4
** Significant at the 1% level; * significant at the 5% level.			

Table 12: Robustness Checks

6 Conclusion

This paper considers how many quality levels a duopolistic firm exports to a given market. Products are differentiated in two dimensions, variety and quality, both of which are fixed to emphasize that a firm selects from its existing product line when choosing the products to export to a small foreign market. I consider only those parameter value for which demand is market-wide.

The firm makes its export decision in two dimstions: how many quality levels to export, and which quality levels to export. If it exports only one, it chooses the one that is more differentiated from its competitor's quality. If the firm has a sufficient tariff preference or if quality differentiation is not too great, then exporting both products will be optimal. If qualities are close enough, then the firm can benefit from having a "buffer product," which allows it to challenge its competitors for market share. But if qualities are too far apart, then there is no need for a buffer product. Mexican data from 1997 to 2008 provides empirical support for the effect of tariff levels on proliferation of the versions of a car model. The evidence on the effect of quality differentiation, however, is mixed.

The analysis in this paper suggests a number of additional directions for inquiry: extending the analysis to encompass more of the parameter space (i.e. to consider cases when demand is not market-wide), permitting each firm to have multiple (fixed) quality levels from which it must choose the appropriate number, and allowing some consumers to choose to purchase nothing (choose an outside option). The empirical results can be extended by analyzing the effect of tariffs and the degree of quality differentiation on market shares as well as estimating the gains from trade to both firms and consumers when trade liberalization leads to product quality proliferation.

Data Appendix

Data	Source	Notes
Car models and versions	Asociación Mexicana de Distribuidores de Automotores A.C. (AMDA)	Some dealers may not be association members; also identifies segment and price
Specifications, per version and model	Edmunds.com	
	automotive.com;	
	motortrend.com;	
	auto-types.com;	Used for cars not sold in U.S. market
	autoevolution.com;	Used for cars not sold in U.S. market
	Reforma newspaper;	Used for cars produced and sold only in Mexico
	online used car listings	Used for cars produced and sold only in Mexico
Fuel efficiency, per version and model	U.S. Department of Energy	
	auto-types.com;	Used for cars not sold in U.S. market
	autoevolution.com;	Used for cars not sold in U.S. market
	Reforma newspaper	Used for cars produced and sold only in Mexico
Applied tariffs	World Integrated Trade Solusion (WITS)	
Gas prices at the pump	German Agency for Technical Cooperation	Only biannual data are available; price in a given year is used for both the current and previous year
Country of assembly	Corporate websites; newspapers and trade journals	
Distances	CEPII	

Table 13: Data Sources

	Truncated Poisson Distribution		
	Robust SE	Clustered SE (model level)	Clustered SE (brand level)
Tariff Rate	-.87**	-.87*	-.87
	(.269)	(.435)	(.55)
Year FE	yes	yes	yes
Brand FE	yes	yes	yes
Constant	-14.9	-14.9**	-14.9**
	(2.16)	(5.64)	(1.05)
Observations	1235	1235	1235
ln(L)	-3759.2	-3759.2	-3759.2

	Truncated Negative Binomial Distribution		
	Robust SE	Clustered SE (model level)	Clustered SE (brand level)
Tariff Rate	-1.10**	-1.10**	-1.10*
	(.211)	(.342)	(.505)
Year FE	yes	yes	yes
Brand FE	yes	yes	yes
Constant	-18.1**	-18.1	-18.1**
	(7.08)	( )	(0.76)
Observations	1235	1235	1235
ln(L)	-3064.6	-3064.6	-3064.6
alpha	.42**	.42**	.42**
(SE)	(.03)	(.06)	(.08)
** Significant at the 1% level; * significant at the 5% level.			

Table 14: Comparing Different Distributional Assumptions

	Engine Size	Horsepower	Physical Size	Wheelbase	Cost per Mile	Fuel Efficiency
Engine Size	1.00					
Horsepower	.89	1.00				
Physical Size	.57	.47	1.00			
Wheelbase	.57	.46	.90	1.00		
Cost per Mile	.71	.80	.46	.44	1.00	
Fuel Efficiency	-.79	-.79	-.59	-.56	-.77	1.00

Table 15: Correlations of Car Characteristics

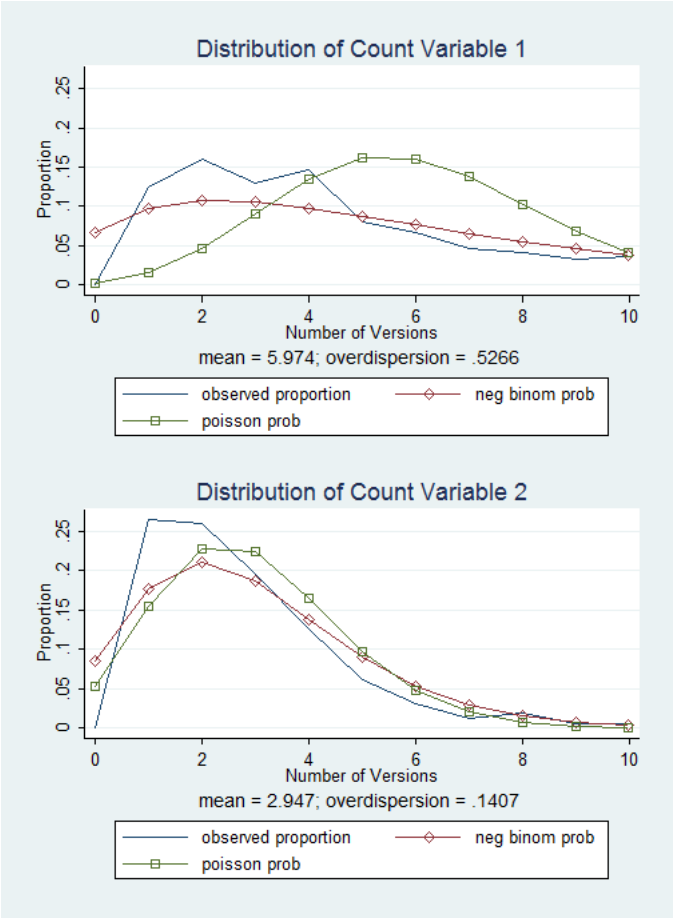

Figure 2: Distribution of Count Variables
Compared to Poisson and Negative Binomial Distributions

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